

Level 1 Functional Diagram — PD-RPM Remote Care System

A function is **what the system does**, not what it is, and is written verb-noun so that the physical solution stays open for the trade study. Figure 1 places the twelve Level 1 functions of the PD-RPM remote care system inside the context diagram established earlier in the semester: inputs enter from the left, outputs leave to the right, controls act from above, and enablers support from below. Each arrow is labeled with the **noun** that passes between functions, and the output of one function is the input of the next.

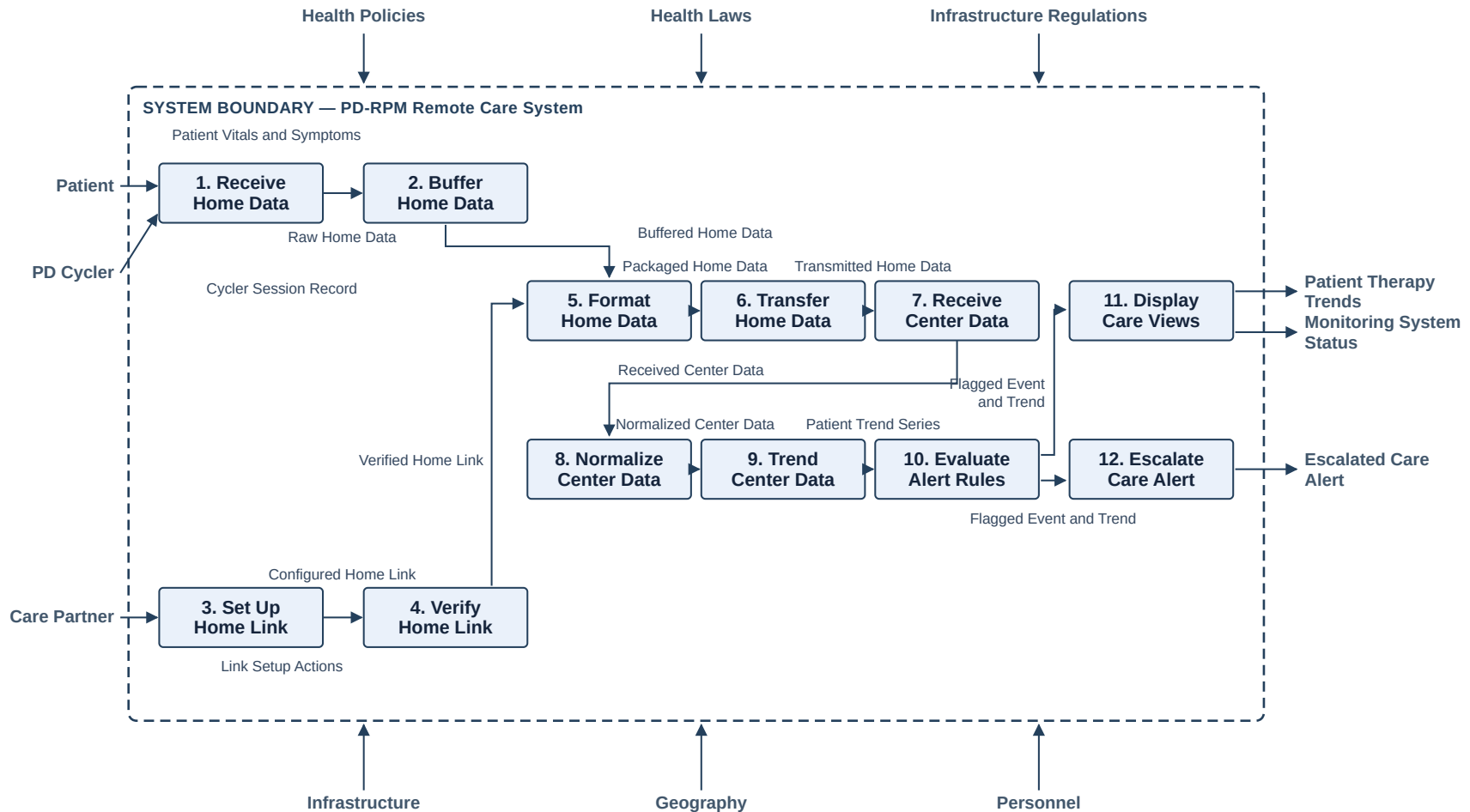


Figure 1. Level 1 functional diagram, drawn as a functional overlay on the PD-RPM context diagram. Twelve verb-noun functions; every flow labeled with the noun that passes.

N² Diagram — PD-RPM Remote Care System

The N² diagram carries the same twelve functions on its diagonal and the same nouns in its cells. **Outputs leave a function horizontally along its row; inputs enter vertically down its column**, so the cell at row 8 and column 9 holds what function 8 delivers to function 9. External inputs and outputs are carried in the flanking columns.

External inputs	1	2	3	4	5	6	7	8	9	10	11	12	External outputs
Patient Vitals and Symptoms (<i>Patient</i>) Cycler Session Record (<i>PD Cycler</i>)	1. Receive Home Data	Raw Home Data											
		2. Buffer Home Data			Buffered Home Data								
Link Setup Actions (<i>Care Partner</i>)			3. Set Up Home Link	Configured Home Link									
				4. Verify Home Link	Verified Home Link								
					5. Format Home Data	Packaged Home Data							
						6. Transfer Home Data	Transmitted Home Data						
							7. Receive Center Data	Received Center Data					
								8. Normalize Center Data	Normalized Center Data				
									9. Trend Center Data	Patient Trend Series			
										10. Evaluate Alert Rules	Flagged Event and Trend	Flagged Event and Trend	
											11. Display Care Views		Patient Therapy Trends Monitoring System Status
												12. Escalate Care Alert	Escalated Care Alert

Alignment check. All 11 internal flows in Figure 1 appear once in the matrix, and the three external inputs and three external outputs appear in the flanking columns. The empty cells are a result in their own right: the matrix is almost strictly upper-triangular, which says the system is a one-way pipeline with no function feeding anything upstream of itself — every loop in this system closes through a clinician, outside the boundary. Controls and enablers act on all twelve functions and are shown in the frame of Figure 1 rather than repeated in every row.